BUDZRESERVE: AN ONLINE RESERVATION WITH BADMINTON RACKET RENTAL, DIGITAL RECEIPT AND QUEUEING APPLICATION FOR BUDZ BADMINTON COURT

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A capstone project outline submitted to the faculty of the Department of Information Technology, College of Engineering and Information Technology, Cavite State University, Imus, Cavite in partial fulfilment of the requirements for the degree Bachelor of Science in Information Technology with Contribution No._____. Prepared under the supervision of Mr. Carlo Malabanan.

INTRODUCTION

Life is flying by these days, but still technology remains a central force in transforming the administration and delivery of services. Digital reservation, queueing and rental systems is among the several improvements software has brought, leading to a much more simple and beneficial process. It includes collecting reservations, checking the court time and date accessibility, approving reservations, address payment methods, keeping accurate customer records, keeping track of the record regarding the reservation and rental processes, and manages fair matchmaking for the customers and more. A working reservation system is vital for resource allocation, conflict mitigation and user satisfaction (Faliha, 2021).

And yet the Budz Badminton Court continue to manually manage reservations, queueing, and racket rental through handwritten records or spreadsheets. At the same time these methods once worked well, but they are now ineffective, error prone, damage risk, and inefficient, particularly at a large scale. Their administration exposes

them to the difficulties of keeping track of court and badminton racket availability, double bookings, loss of racket due to the damage of records, and communicating about cancellations in a timely manner, resulting to frustration for staff and customers, and not meeting expectations of service in the modern era specially they don't provide any receipt. Aside from the manual reservations and racket rental, the other problem they want fix is the queueing process, because it plays a crucial role in their business, a queueing is a process in which a group of players or club reserves dozens of courts in the facility. The Club is particularly run by what they called a Q-master (Queueing master), he/she is the one who organizes the sets or games of the players in the whole reservation time frame. The problems we encounter during our field study are the following; first the manual listing of players, pairing of players, that typically results in longer waiting times and sometimes produce unfair matchmaking for the players.

The proposer´s solution to the earlier stated problem is a system named "BudzReserve: An Online Reservation with Badminton Racket Rental, Digital Receipt and Queueing application for Budz Badminton Court" The main purpose of this system is to use an automation regarding to the reservation, queueing and racket rental. A new centralized online platform enables customers to book courts and racket online, check real time availability and receive a digital receipt after the reservation process. In addition, it allows administrators to control their reservations more straightforwardly and, therefore, ensure smooth operations in the improved service delivery. It shows that the Reservation and Queueing System will remove the slow, manual and time-consuming process that normally found in traditional methods and use an automation to reservation system and thus, will reduce errors and increase customer satisfaction. With this cutting-edge solution in place, Budz Badminton Court will contribute to enhancing the quality of their services, considering the increasing demand for available and dependable reservation systems.

Project Context

On March 12, 2016, the Budz Badminton Court (BBC) in Bacoor, Cavite opened its doors. Starting with just six courts, the facility quickly became popular with local badmintonists. As a result of high demand, Budz Badminton Court (BBC) then expanded and added another three courts in 2019. After another renovation, four extra courts were added and they upgraded all the flooring to taraflex in 2022, bringing their court tally up to 13. As customer and facility size grows rapidly, manual reservations, equipment rental management, and the queueing done by staff and q-master have become increasingly insufficient in enabling them to do so quickly and accurately. Multiple customers chase slots during peak times, court availability, and queueing management are one of Budz Badminton Court (BBC) biggest problems to handle and monitor.

The manual method is often prone to errors, time consuming, and having multiple reservations in the same time slot is very inconvenient for customers and staff. A third recurring problem has been keeping accurate, organized records. Handwritten documents or spreadsheets being misplaced and booking details going missing can bewilder and frustrate customers. Specially they don’t provide a transaction receipt slip.

Objectives of the Project

The main goal of this study is to develop BudzReserve: An Online Reservation with Badminton Racket rental, digital receipt and Queuing application for Budz Badminton Court, providing a more efficient, user-friendly, and provides automation way to handle court reservations, racket rental, and queueing management and receipts. It specifically aims to:

1. Enhance the reservation process and improve user experience by:
    a. Automating the reservation, racket rental and queueing process with an intuitive online system that allows customers to reserve courts, equipment, and manage queueing anytime and anywhere.
    b. Ensuring fast, accurate, and secure transactions through an integrated email notification system, providing instant reservation confirmation and digital receipts.
    c. Implementing a real-time availability feature to prevent double reservation and improve court scheduling efficiency.
2. Eliminate human errors in manual reservation and receipt handling by:
    a. Digitizing data entry to ensure accurate and consistent customer records.
    b. Automating reservation confirmations and receipts through email notifications, reducing the risk of lost or misplaced receipts.
3. Optimize court utilization and revenue management by:
    a. Implementing an availability tracking system that monitors court usage and provides insights on peak hours and customer preferences.
    b. Developing a reporting and analytics feature to help administrators track revenue, peak booking times, and customer for better business decisions.

To develop BudzReserve: An Online Reservation with Badminton Racket Rental, Digital receipt, and Queueing application for Budz Badminton Court, a combination of hardware and software is essential for efficiency and security. The system will use MySQL for database management, PHP for server-side processing, and XAMPP as the local development environment. The front-end will be built with HTML, CSS, and JavaScript, ensuring a user-friendly interface, while Visual Studio will serve as the primary IDE. Hardware requirements include a dedicated server or cloud hosting, stable internet access, and desktop or laptop computers with sufficient processing power.

After developing BudzReserve, a structured testing and evaluation process will ensure its functionality, reliability, and security. The system will undergo unit, integration, usability, performance, and security testing to verify seamless operation. User feedback will be collected, and necessary optimizations will be made before final validation.

The implementation will follow three phases: deployment, training, and maintenance. Deployment involves setting up the system, migrating data, and conducting live tests. Training will be provided to administrators and staff, while customers will be informed through digital marketings. Finally, continuous monitoring, updates, and customer support will ensure system efficiency and long-term usability. This structured approach guarantees a smooth transition to an automatic and user-friendly reservation system for Budz Badminton Court.

Purpose and Description

The project will be conducted in order to make the BudzReserve more user-friendly, efficient, and well-organized. Users will have online access to check and reserve courts, and the administrator will have a digitalized and compressed record of customers and transactions while any Q(master) can manage queueing through the website and application.

1. This system provides a quick and efficient approach to managing the court reservation, queueing, and racket rental process for instant task execution.
2. The new and improve system would allow users to select and reserve any available courts and racket.
3. Once a court is reserved, users will receive an email with booking and payment details for confirmation so they know exactly what reservation they have made.
4. Real-time updates on reservations and availability, customer data, are easily managed by the admins.
5. Uses automation in queueing process.

The study proposed concerns Budz Badminton Court which still use manual booking process that is spreadsheet-based and handwritten. Maintenance of records in this manner is not a very accurate, efficient and productive process, as it is manual, prone to errors and time-consuming.

Once BudzReserve: An Online Reservation with Badminton Racket Rental, Digital Receipt and Queueing application for Budz Badminton Court is implemented it will give the following significance particularly to the following:

1. Client – it will be beneficial for them through the way the system will provide accurate reservation and transaction history data. It will maximize the operating time of the company and make employees more productive that can make more sales and income.

2. Customer- It will be beneficial for the customer because it will save them more time in the reservation process. Preventing them from getting frustrated.
3. Proponents – it will help them accomplish and meet the requirements in order to pass the subject.
4. Future proponents – through this research future proponents will have an idea and additional references in order for them to accomplish documentation they need to fulfill.

Time and Place of the Study

The study will run from May 25 to December 14 in Cavite State University (Imus Campus), and during this time system design, prototype creation, user trials, and data evaluation will all be possible. The proponents will ensure that all the phases of the project from the preliminary setup to the presentation are treated in a way that ensures that only the necessary objectives are accomplished within the time available. The early stages of the project will focus on system development and functionality (operational) testing, with subsequent phases centering on extensive trials (for system performance testing).

Scope and Limitation of the Study

The primary goal of this research is to create and implement the BudzReserve: An Online Reservation with Badminton Racket Rental, Digital Receipt and Queueing application for Budz Badminton Court. The system hopes to improve the process of reserving, racket rental and the queueing process in the Budz

Badminton Court, for potential customers by providing them with a digitalized system and the ability to schedule court times and receive digital email receipts for their transaction. It will be all possible by providing the following modules;

To guarantee effective operation and a flawless user experience, the system is made up of multiple essential parts. The Payment Module supports safe transactions and keeps clear payment records, the Reservation Module permits real-time court bookings and lets users modify their reservations. To keep consumers informed, the Notification Module automatically sends out email receipts and booking confirmations. While the Racket Rental Module this module enables users to view the list of available badminton racket, check their availability, and submit rental requests.

While the Reporting Module creates daily summaries of reservations and money activity for performance monitoring, the System Administration Module handles user accounts, permissions, and court information for administrators. Profiles, reservations, and payment histories are among the crucial data that is safely arranged by the Database Management Module. The Integration and Security Module additionally establishes connections with third-party systems, such as payment gateways, and guarantees data security by implementing access controls and encryption. Finally, Budz Badminton Court's efficiency is increased by automating the manual queueing procedure using the Queueing Management Module.

The people involved in the data gathering and data evaluation for the improvement of the system are the following:

- Owner/Businessman/Client
- Employees

Convenience sampling technique is the method used for gathering relevant data and information, as it is giving convenience not only to the proponents but also to

the participants, whenever and wherever they are available. Standard deviation is the statistical tool that is deemed optimal for the evaluation of data gathered.

The system currently has the following limitations:

1. Internet Dependency, the system requires a stable internet connection for users to access the platform and for email notifications to be sent successfully. Any internet outage may disrupt the reservation and notification process.
2. System Downtime refers to periods when the BudzReserve system becomes temporarily unavailable due to technical issues, updates, or external factors. This can significantly affect user experience and business operations.
3. Does not include such sophisticated features, such as analytics tracking of reservation trends.
4. The payment method “paymongo” can’t be used when the API is down.

However, these shortcomings could be addressed in future versions, to enhance the system's capabilities and go beyond some of the existing requirements.

Conceptual Framework

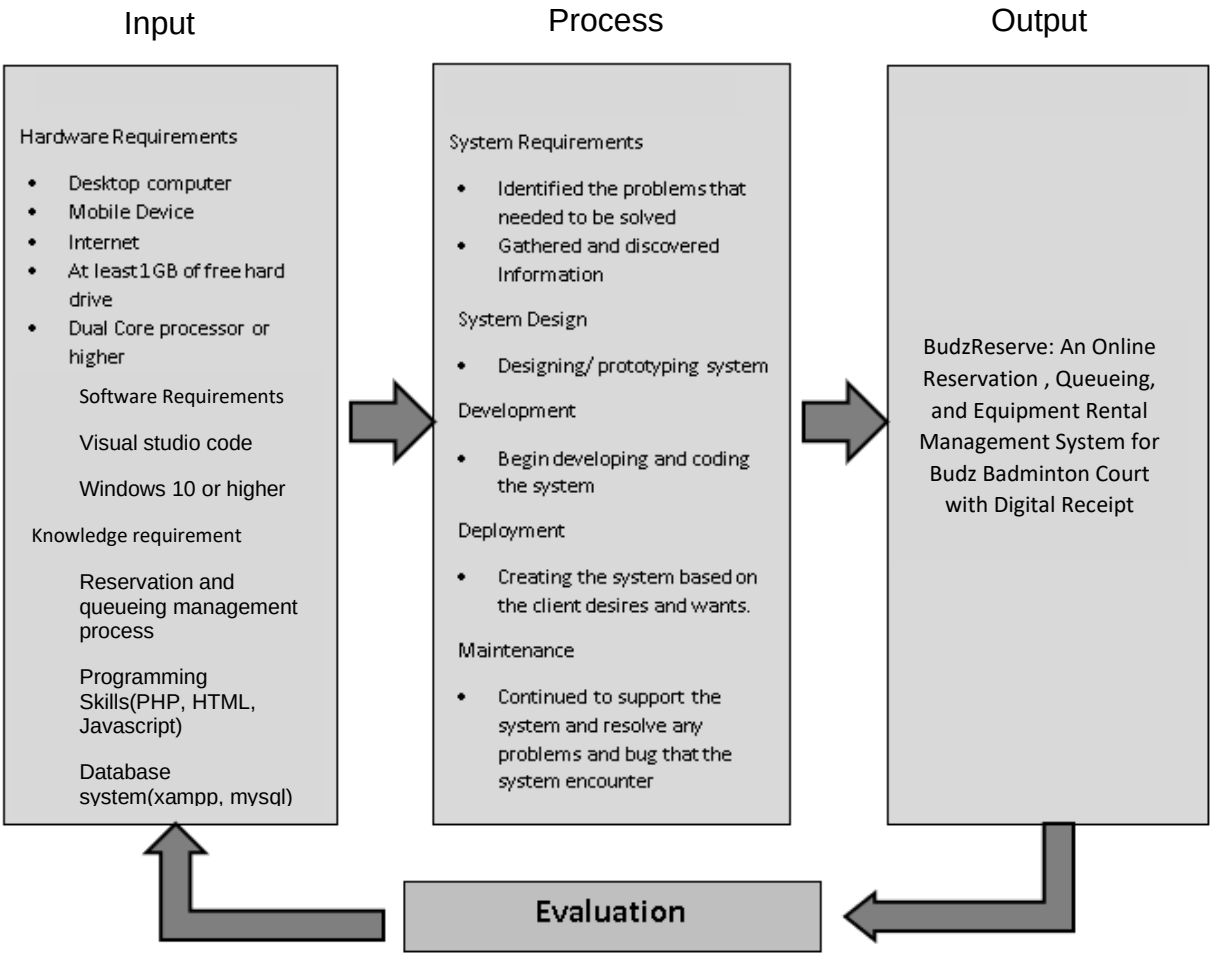

Figure 1. Conceptual framework

Figure 1 shows the study's conceptual framework of the project, where they must input their knowledge of the reservation management system process, as well as their skills in programming, and database systems. In terms of software, they need at least Windows 10 and above, as well as Php, javascript and Xampp for development tools. In terms of hardware, they need a desktop computer, at least 1GB of free hard drive space, and a dual core processor or higher. System requirements, system design, development, deployment and maintenance are all part of the process. The proponents identified the problem to be solved as well as gathered and discovered

information during system requirements. In addition, the design will be created by the proponents in System design. The proponents will begin the system's development and coding in the Development phase. The developers will change, add, or remove features based on the client's preferences during the Deployment phase and finally, in Maintenance, the proponents continue supporting the system and resolving any bugs or problems that arose.

As a result of the research, BudzReserve: An Online Reservation, Queueing, and Equipment Rental Management System for Budz Badminton Court with Digital Receipt

The final statement refers to the evaluation phase, which is designed to determine the suggested proposed system's availability, usability, reliability, and security. It must be validated to see whether or not the proposed system fits all of the requirements.

Definition of Terms

Administrators - Individuals tasked with planning, organizing and overseeing institutions, facilities or systems. They maintain operations, fix issues, and make sure everything runs smoothly in their domain.

Badmintonists - players of badminton, either in competition (often, professional) or as recreation. They are competing in a racket sport that requires speed, precision and anticipation.

Email notification - to send a pre-authored text based receipt to email.

Online reservation and Queueing system – A website that specifically manages reservation and queueing process.

Prototype - an initial or a working model of a system or product for checking its feasibility, function or design. Another use of the synthetic data is evaluation and improvement before final production.

Queueing – A term used in badminton in which a set of random players play and compete with each other randomly.

Q-master (Queueing master) – It is a term used by badminton players, in which the one who manages and organizes the queueing process in the badminton court.

Software - is a set of data, programs, or instructions that operate machines and perform specific tasks. They include utilities, operating systems and apps that allow gadgets to perform tasks such as process data, communicate and entertain.

Spreadsheets - are digital apps that organize data into rows and columns, such as Google Sheets and Microsoft Excel. Typically, They involve formulas as well as charts and filters and are used for computation and data processes.

System - an interconnected structure composed of numerous constituents, processes, or elements that work in unison to fulfill a specific function. It can include operational workflows, organizational charts, or technical architectures.

Taraflex - A specified brand for synthetic floor covering used for indoor sports. High quality: Makes use of for its durability, shock-absorbing properties, and enhancement of athletic performance with lower injury risk.

REVIEW OF RELATED LITERATURE AND STUDIES

Following a thorough search by the researchers, the researchers present the relevant literature and studies following an extensive search. Additionally, discussions on the facts pertinent to the research are included.

Review of Related Literature

To combat such inefficiencies like making manual bookings and overlaps, Lim et al. (2023) Developed a badminton court reservation system which runs over the web for Batu Pahat (2023) The system contains reservation, user administration, court management, and report generation modules and covers ten courts. With its foundation rooted in the Agile methodology, this system guarantees that specific criteria are fulfilled and provides the admin, users, and court owners with the tools to manage reservations efficiently. This software allows badminton court owners to help them avoid the mistake and maximize the rental besides provide customers a hassle-free experience of online booking.

DKurniawan, Richi (2024) improved the speed of payment transactions on a website renting a badminton field by implementing the Midtrans payment gateway and employing the Paired T Test to examine the results. It addressed the issue of slow transactions during booking to enhance the user experience. The results demonstrated the efficiency of technology to accelerate services in the sports rental market that offer improvements and the speed of rent transactions.

Ahmad et al. (2024) developed the Sport Facilities Reservation System (STFS) for Universiti Tun Hussein Onn Malaysia (UTHM) to improve the utilization of

university sports facilities. The system features user registration, facility booking, and database management, built using MySQL, Xampp, CSS, and PHP. STFS aims to streamline the reservation process and support the university's role in community services. Expert evaluations revealed that while the system had some weaknesses, it successfully met its goals, providing a more efficient and accessible platform for reserving UTHM's sports facilities.

In order to replace manual procedures that are prone to mistakes and inefficiency, Febriansyah et al. (2023) created a mobile application to expedite sports facility leasing in Indonesia. The app improves facility management accuracy and efficiency by allowing managers to monitor and print rental reports and customers to book fields with ease using a REST API and the Laravel framework.

Review of Related Studies

Foreign study

Utilising its player chat, real-time availability and notifications, Radzuan (2024) developed an iSmashON smart phone application to help find and reserve badminton courts. Player feedback (N=30) was positive and highlighted the effectiveness of the technology along with ease of use according to the Technology acceptance model (TAM). It aims to improve the booking experience of the badminton community.

Ab Ghani et al. (2023) presented a sports center booking system, to help improve productivity, using WordPress, XAMPP, MySQL, PHP and HTML to solve issues in existing systems, such as maintenance, and booking confirmations. The system also integrates with Stripe payment, along with registration, payment, booking

availability, and QR code generation for confirmation. Students and staff members check in using ID, and new members sign up and pay fees. Bookings are confirmed by administrators, who send out an email notification with a QR code. The new system increased booking efficiency with 86.66% more usage through the old system.

Jafridin et al. (2023) The Pura Kencana Sports Centre Management System is developed to solve problems of the paper-based booking system, causing customers to experience repeated bookings and lose clients' data. Built on PHP, MySQL, HTML5, and ZF2, the system simplifies administrative processes and simplifies court reservations. The system implements Agile Model to ensure a structured booking process, of aiding data management as well as a future integration with the payment gateway to cater to both the users and the sports facility management.

Local study

German et al. (2024) developed a systematic approach to address the inefficiencies of manual sports facility reservations in the Philippines. developed an online booking system called "Appoint Deportes." The technology allows users to book online, track real-time availability and pay at machine-freed convenience. By leveraging expert feedback and flowcharts, the team laid out user-friendly interfaces that save costs, enhance efficiency, and encourage environmental conservation—points that underscore the importance of introducing tech solutions to streamline booking processes and elevate user experience.

Pedroche et al. (2024) designed a system that improves the reservation and time monitoring system of INSPIRE Sports Academy (ISA) in the Philippines. Staff monitoring and email reservations were previous manual techniques, and they failed. The new system included QR Code scanning for reservations, streamlined entry/exit

tracking and automated services to help improve operational efficiency. By employing the Iterative Life Cycle spiral loop model, the team was able to get client feedback and integrate it, thus creating high automation that enhanced service convenience and reduced the existing complex process.

System Technical Background

Michael Plainer (2021) With more than 11 million users, Microsoft's Visual Studio Code (VSCode) is a popular code editor. Although it is based on open-source code, or OSS, Microsoft adds proprietary features like marketplace integration and makes it non-free by releasing it under a unique license. Without the proprietary features, other distributions like as VSCodium provide a free version of Code-OSS. Additional analysis of the open-source features of VSCode and Code-OSS is provided by an initial study.

According to ASHWANI K (2021), XAMPP is an open-source web-solutions kit that provides Apache delivery for a variety of servers and command-line executables, as well as Apache API, MariaDB, PHP, and Perl modules. XAMPP can be validated on computers and laptops with the help of a local host or server.

Ebrahim (2024) This study measured the effectiveness of an interactive instructional website (designed in HTML, CSS and JavaScript) on the learning performance of students from the Al-Azhar University Teaching Technology course. The study consisted of twenty fourth-year students, ten for the experimental group and ten for the control group. An exam to measure participants' knowledge acquired and an electronic mental ability test were included to assess the impact. These showed that the students learnt better with website ease of use and instant feedback.

The study suggested the provision of a guide to help students make use of interactive instructional websites.

Synthesis

The studies analyzed showed the importance of technology and its role in changing the reservation systems of sports facilities for an efficient process, in terms of users and operational process. Systems such as those by Lim et al. (2023) and Ahmad et al. (2024) eliminate human mistakes and ensure you get the most out of your facility through features like instant booking, simple user interfaces, and secure payment gateways. As mentioned by DKurniawan (2024) and Ab Ghani et al. and (2023), further enhance transaction speed and reduce administrative costs. Local analysis, among others by German et al. (2024) and Pedroche et al. (2024)—shows successful applications of automation in the Philippines emphasizing real-time tracking, QR codes, and iterative development for customized solutions. User feedback, agile development, and use of automation are key to ensuring continuous improvement and usability as per the studies globally and. locally. However, the conduct of real-time reservation, safe payment method, automatic alert, and data provision to the users help to join the combination of operational efficiency and end-user satisfaction. With this, future systems have the potential to offer scalable, efficient and user-friendly sports facility management solutions leveraging the best practices of studies carried out on global and local front.

Offering email receipts, API payment integration and accommodating the special demands of badminton court reservations, we expand on prior developments to show "BudzReserve" ingenuity within the industry.

METHODOLOGY

This chapter presents the materials, methods, and data gathering techniques, data analysis, and implementation plan.

Requirement Analysis

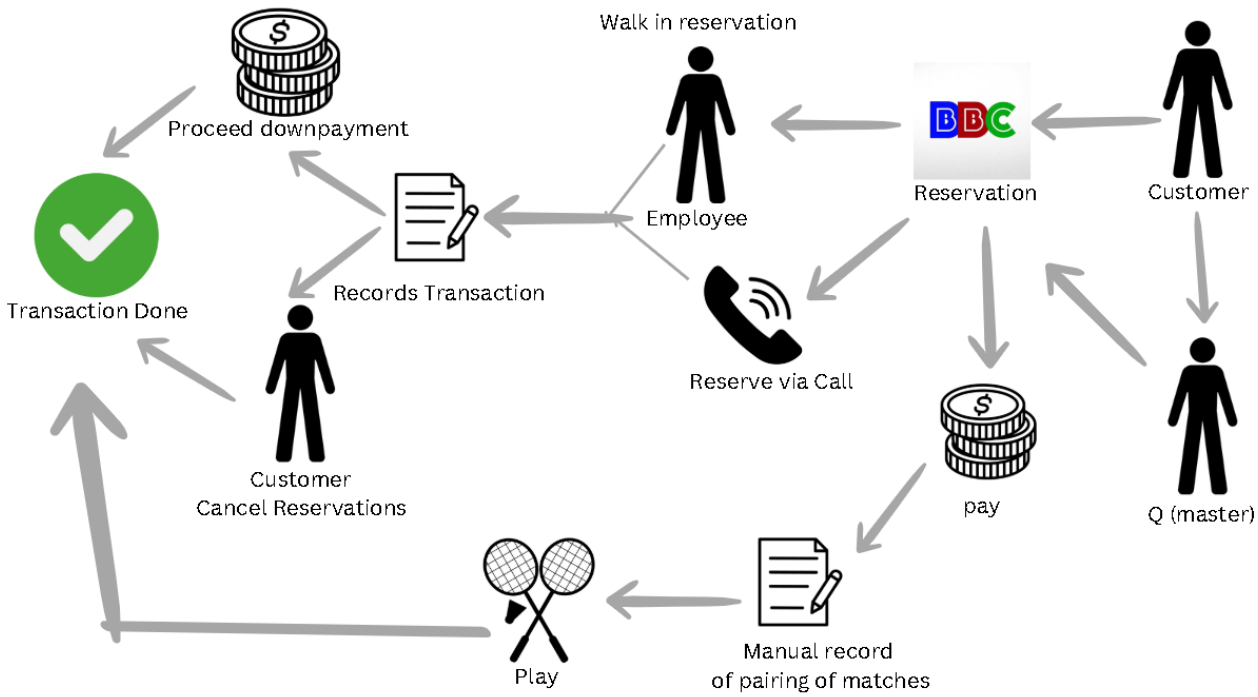

Figure 2. Manual Process

Figure 2 shows how Budz Badminton Court (BBC) handles a daily basis manual transaction. The transaction starts, when the customer decides what type of reservation is he/she gonna do, through walk in or Via call the equipment rental process is also process by that two option. After the customer decides the employee will ask for the customer details ang manually records it on the paper. And then the customer can decide if he/she wants to cancel the reservation or proceed with the down payments. So, the Budz Badminton Court uses a downpayment method in order for them to prevent getting tricked or scammed.

Requirement Documentation

Based on the interview conducted with the client, the following features have been agreed:

Table 1. Features of the proposed system

Main Features	Detailed Features
User Registration and Login	1. Enable users to register an account using personal information and a secure login password. 2. Enables users to modify their profiles.
Real-Time Court Reservation	1. Present free-place open time slots as they are up available. 2. Book a date and time for a reservation, with recommendations. 3. Payment ones using their online gateway
Payment Integration	1. Allows payment to be made securely at the online gateway. 2. Indicate that a payment transaction was accepted into the database.
Email Receipt/Notification	1. Confirmation email which is sent automatically 2. Confirmation and the updated information for the court booking and payment fixedly.
Admin Monitoring Dashboard	1. Provide admin a real time updates 2. Manual updating of reservations, can be canceled
Daily Reservation Report	1. Daily report and email about the status of reservation and payment. 2. Highlight user data and summarize the profits obtained.
Queueing page	1.Using automatic matchmaking for the players. 2.Categorize the level of the player and generates a fair game for each players 3. using automation to maximize the time and increase profit.
Equpment Rental Page	1.Show all the equipment that is available 2.The user can select and rent a equipment just like the reservation process

The following table, however, presents the non-functional requirements of the proposed system:

Table 2. Non-functional requirements of the system

QUALITY ATTRIBUTES	DESCRIPTION
Availability	The system should support up to 100 registered users and handle up to 150 reservations per day. Furthermore, it must be adaptable enough to accommodate new features, such as the ability to handle several courts, without needing large changes.
Usability	The system should be user-friendly, allowing users to make bookings quickly and easily. Clear instructions and mobile optimization are required to guarantee it works properly on both PCs and smartphones.
Security	User data, including login and payment information, must be stored securely using encryption technologies. Access to certain features should be limited to authorized users, and all actions should be documented for auditing.
Scalability	It should also be able to handle mistakes gracefully, delivering clear error messages without crashing. Using automation in email receipts must be trustworthy, ensuring that users receive their confirmations without issue.

Software Design

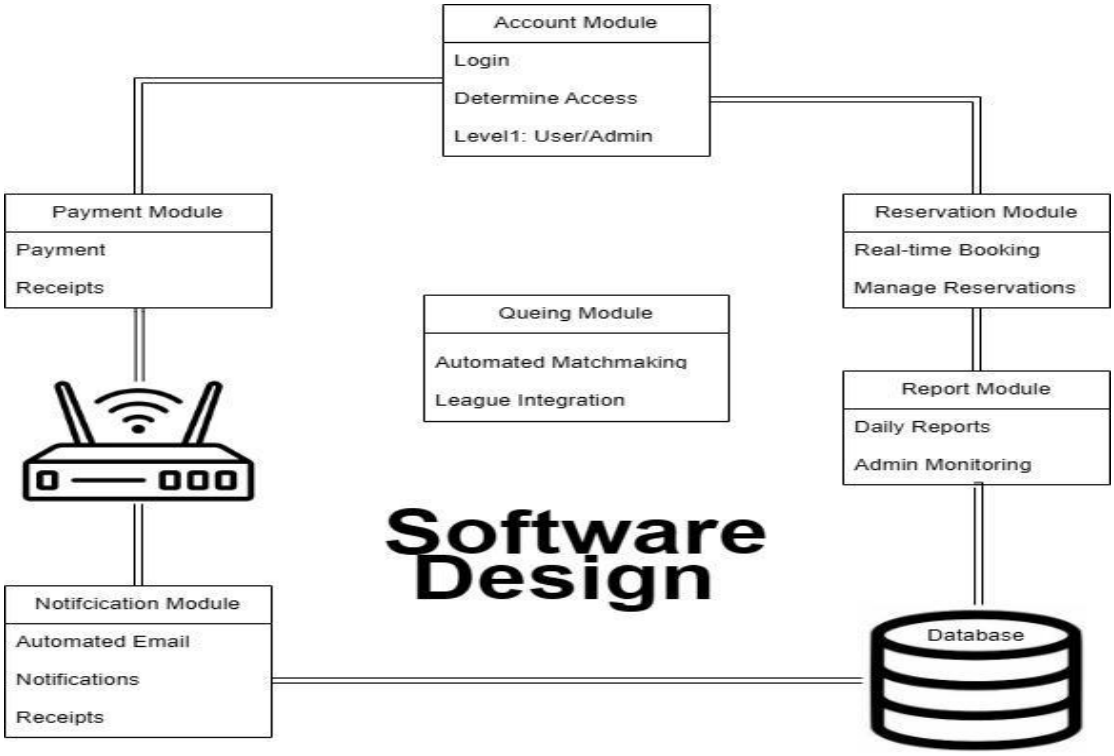

Figure 3. System architecture

The system architecture, presented in Figure 3, is composed of different modules:

Account Management Module:

The Account Management Module allows users to create and manage their personal accounts easily. Users can sign up, log in securely, and update their profile information at any time. It also includes password recovery features and lets users view their booking history and transaction records, helping them keep track of their activity on the platform.

Reservation Module:

The Reservation Module enables users to reserve badminton courts in real-timeUsers can simply check the court's availability and make bookings right away. It also includes features for managing current reservations, allowing users to change or cancel them as needed.

Equipment Rental Module

The core of the system, this module enables users to view the list of available sports accessories and equipment, check their availability, and submit rental requests. Admins can approve, reject, or update the status of the rentals (e.g., issued, returned, damaged).

Payment Module:

The Payment Module integrates secure payment processing into the system, allowing users to conduct reservation transactions quickly and securely. Furthermore, this module records all payment transactions to maintain openness and accountability.

Notification Module:

The notification Module is one of the important features of the system that the researchers want to integrate, notification module is responsible for sending email notifications to the users, ensuring timely information on the progress of booking of the user.

Reporting Module:

The Reporting Module generates daily reports summarizing reservation activities and financial transactions. These reports offer administrators valuable insights into system utilization, assisting them in monitoring performance and making educated decisions.

Database Management:

Database Management pertains to the storage and oversight of all essential data within the system. The database Management module contains profiles, reservation information, payment history, system logs, and other necessary information of the system.

System Administration:

The System Administration module enables administrators to handle the system successfully. It contains facilities for adding and deleting user accounts, as well as modifying permissions. Furthermore, administrators may monitor and manage court information to guarantee correct availability and maintenance scheduling.

Integration and Security:

Integration and Security focus on integrating with external systems, such as payment gateways, to enhance functionality and user experience. Security measures, including encryption and access controls, are implemented to protect user data and ensure system integrity against unauthorized access and breaches.

Development and Testing

Project Development

The program will be developed by the researchers using the spiral process as a guide. The spiral technique ensures a well-structured and adaptable process by combining iterative development with a heavy emphasis on risk assessment, making it extremely effective for the suggested system. Planning, risk analysis, engineering, and assessment are the four (4) primary stages of this paradigm, and they are repeated in a sequence of spirals until the system is completely created. In order to enable continuous improvement throughout the development process, each iteration entails

enhancing requirements, creating and constructing prototypes, testing, and analyzing feedback. The spiral methodology's iterative structure guarantees that hazards are recognized and dealt with early on while producing a top-notch solution that is customized to meet user needs. Figure 4 depicts the phases of the spiral approach.

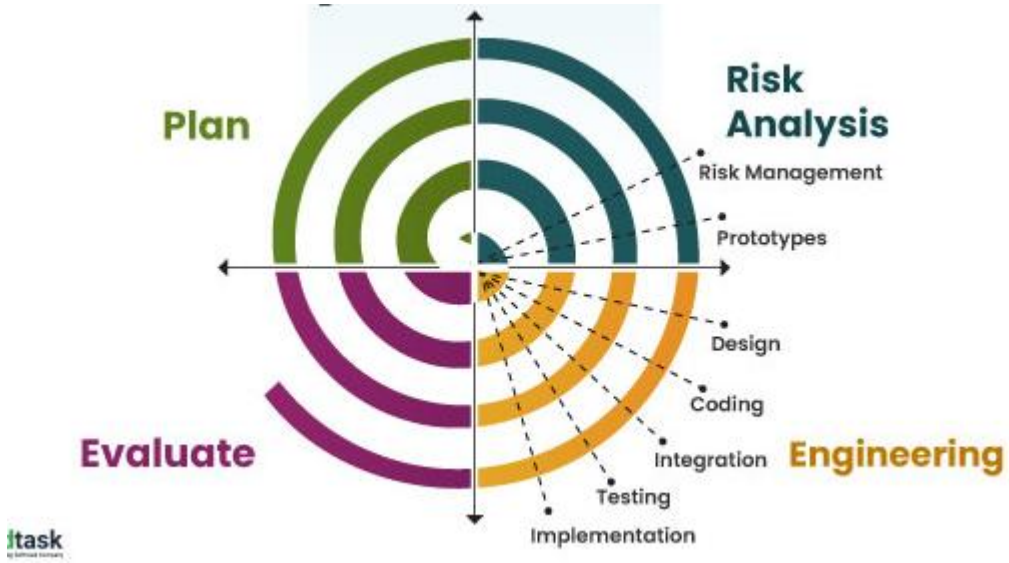

Figure 4. Spiral Methodology

Planning Phase

In order to determine the issue that the system seeks to address, the researchers plan extensively throughout this phase. In order to guarantee a complete grasp of the intended functionality, they collect comprehensive needs through user consultations and discussions. The goals of the system and the parameters of its development are the main emphasis of this phase.

Risk Analysis Phase

Potential dangers are recognized, evaluated, and reduced during this stage. To test particular parts or features, the developers build prototypes, or early iterations, of the system. To make sure they meet user expectations, these prototypes are shown to the client for input. As needed, changes are done to improve the functionality and design

Engineering Phase

The system is developed gradually, with each iteration building on the one before it, based on the improved design. Using programming languages and techniques that are suited for the system requirements, programmers create the code. Prototypes and designs are transformed into working parts that make up the entire system.

Evaluation Phase

Every iteration is put through a rigorous testing process to make sure it satisfies the set requirements and operates as intended. To make sure that every component functions as intended, developers test for errors, inconsistencies, and defects. This stage incorporates the client's input to confirm the system's operation.

Project Testing

A. **Unit Testing** – All system modules are tested one by one to ensure that all functions within each module are functional. Starting with the most basic and first module, the user login form, followed by the system's main functions.
B. **Integration Test** – After the unit testing, another test will be performed, which is the Integration Test. During this phase, modules are tested in pairs to ensure that they work when combined with the other module. Modules should work together to ensure that data is properly executed.
C. **System Test** – This is the overall system testing. An overall system test should be performed to ensure that the two tests are done properly. To avoid delays

in the implementation phase, the system should be free of all system errors during this test.

Data Analysis Plan

The researcher personally conducts research by interviewing people in order to gather information for the system's development. The researchers collect data on various materials relevant to the study that is being conducted. The information gathered during the interview with the business owner is chosen by the researcher to aid in the development of the inventory system.

Interview

An interview is essentially a structured conversation in which one party asks questions and the other responds. A one-on-one interview between an interviewer and an interviewee. The interviewer poses questions to the interviewee, who usually responds by providing information.

- During this phase, the researchers conducted an interview with the business owner in order to gather information for the program's development.

Questionnaire

A questionnaire is a type of research device or instrument that consists of a series of closed-ended or open-ended questions. The goal is to gather relevant information from respondents. When you give the respondent the option to provide a longer response, you can gain more insights because they can elaborate on their ideas. Convenience sampling is a type of sampling where the first available primary data source will be used for the research without additional requirements. In other words, this sampling technique

involves getting participants wherever you can find them and typically wherever is convenient. The overall simplicity and ease of research makes it more accessible for researchers (Gall, Borg, & Gall, 1996).

- During this phase, the researchers ask practical answerable based on the problem that they are encountering with their business inventory, as well as questions about the history of their business

The researchers will use a pointing system to acquire the level of agreement of the respondents on the questions in the survey questionnaire. The points to be used are:

Table 3. Options in each item of the questionnaire

SCORE	INTERPRETATION
5	Excellent
4	Very Good
3	Good
2	Fair
1	Poor

Table 2 shows the Likert scale that can be used to generate the overall interpretation of the mean score of each item.

Table 4. Likert scale

RANGE OF WEIGHTED MEAN	INTERPRETATION
4.51 - 5.00	Excellent
3.51 - 4.50	Very Good
2.51 - 3.50	Good
1.51 – 2.50	Fair
1.50 and below	Poor

On the other hand, the following statistical procedures will be used to analyze the data to be gathered from System Evaluation Questionnaire. The results of the statistical procedure will determine the general perception of the respondent on the system.

Sample mean is the average score of a sample on a given variable.

Formula:

$$\underline{x} = \frac{\sum_{i=1}^{n} X_i}{n}$$

Where:

$\underline{x}$ = mean

X_i = representation of each observation from respondents

n = total number of respondents

Sample standard deviation is a measure of the spread (variability) of the scores in the sample on a given variable.

Formula:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (X_i - \bar{x})^2}{n-1}}$$

Where:

$\underline{x}$ = mean

X_i = representation of each observation from respondents

n = number of respondents

s = sample standard deviation

Percentage determines the frequency counts and percentage distribution of personal related variables of the respondents.

Formula:

$$\% = \frac{f}{n} \times 100$$

Where:

% = percentage

f = frequency

n = total number of respondents

Implementation Plan

Budz Badminton Court will review the revised and finalized version of the project" BudzReserve: An Online Reservation and Queueing Management System for Budz Badminton Court with email notification" after testing. The researchers will then transfer the system along with complete documentation if the management chooses to implement it. It will come with a formal letter stating that the system is being freely provided to the organization, and that the researchers will not be responsible for its ongoing maintenance. If the solution gets deployed, the researchers would plan to follow it up with a smooth transition by providing user training sessions and also initial technical support and monitoring of the implemented system to address any immediate issues in the implementation phase.

Strategy	Activities	Persons Involved	Duration
• Introduce the system to the organization	• Present the features and functionalities of the system to the employee and owner.	• Owner and Employee/staff	• 9:00 – 10:00 am
• Conduct system testing	• Guide the employee and owners about the overall process of the system.	• Owner and Employee/staff	• 10:00 – 12:00 pm

Table 5. Implementation plan

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